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The 10th International Conference on Sustainable Urban Development

FOSSIL FUEL POLICIES AND FIRM INNOVATIONS

Evidence from a Developing countries Database

PRESENTATION – 2024

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Motivation

Innovation is a key driver of economic growth and competitiveness, particularly in developing countries where rapid economic changes and market volatility are prevalent. In such environments, a range of factors influences innovation, with economic variables like fuel prices playing a critical role.

In developing countries, fuel prices significantly impact production costs, operational strategies, market dynamics, and overall profitability. This study explores how fluctuations in fuel prices in these economies relate to **firms' product innovation**. Understanding this relationship is crucial, as it can reveal how economic pressures, such as rising costs and market uncertainties, drive firms in developing countries to innovate.

► By examining this relationship, we gain insights into how firms in developing countries adopt new technologies or develop products to cope with economic challenges and sustain market presence. This analysis contributes to our understanding of the unique economic factors that motivate firm-level innovation in developing contexts.

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Research Questions

- What impact do fuel price regulations have on the innovation of firms?
- What are the implications of these interactions for policy development in developing economies?

1. The critical role of fuel prices as a driving force in fostering innovation and shaping the economic dynamics of former socialist countries.
2. Understanding the implications of fuel price regulations is essential. As these countries continue their transformation from centrally-planned to market-oriented economies.
3. Firm-level innovation is vital for economic resilience, particularly in transitioning economies, making it crucial to analyze the impact of regulatory frameworks on innovation.

➔ This research seeks to provide valuable insights that can guide policymakers in designing fuel regulations that encourage innovation while balancing economic stability.



ENTERPRISE SURVEYS



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- ✓ This study utilizes the "Standardized Survey Data" provided by the World Bank, covering 47 developing countries and encompassing 321,500 firm-level observations collected from 2011 to 2015.
- ✓ This data was collected in WBES and IFS.
- ✓ The data set is not a panel, as different firms were surveyed across various waves during the period from 2011 to 2015.
- ✓ Additionally, monthly gasoline price data was sourced from the World Bank and Global databases.

ha14d	ha14m	ha14y	ha14h	ha14min
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HA.0b Same respondent as the ES	ha0b	
YES		1
NO		2

INTERVIEWER READ THE FOLLOWING TO THE RESPONDENT BEFORE PROCEEDING

The goal of this survey is to follow-up on the interview recently carried out about the business environment in Nepal and to collect additional information on innovation.

B. PRODUCT INNOVATION

INTERVIEWER READ THE FOLLOWING TO THE RESPONDENT BEFORE PROCEEDING

I would like to start by asking you some questions about any new or significantly improved product or service introduced by this establishment, where “new” means new to the establishment and not necessarily new to the market. The next questions will refer to this new or significantly improved product or service as an “innovative” product or service.

HB.1	From fiscal year 2009/2010 thru 2011/2012, did this establishment introduce any innovative product or service? SHOW CARD 1
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Yes	1	
No	2	GO TO HB. 19
DON'T KNOW (SPONTANEOUS)	-9	GO TO HB. 19
		hb1

HB.2	From fiscal year 2009/2010 thru 2011/2012, how many innovative products or services did this establishment introduce?
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Yes	1	
No	2	GO TO HB. 19
DON'T KNOW (SPONTANEOUS)	-9	GO TO HB. 19
		hb1

HB.2	From fiscal year 2009/2010 thru 2011/2012, how many innovative products or services did this establishment introduce?
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	Number	
Number of innovative products		hb2 IF 1, GO TO HB.5
DON'T KNOW (SPONTANEOUS)	-9	

WBES SURVEY

IFS SURVEY

VARIABLES

❖ Dependent variable

- 1. Product innovation:** A dummy variable is equal to 1 if a firm introduces a new or significantly improved product/service in the last three years; 0 otherwise. (IFS)
- 2. Process innovation:** A dummy variable equal to 1 if a firm introduces a new or significantly improved process, such as innovative methods of manufacturing products/offering services, logistics, delivery/distribution methods, or supportive activity/process in the last three years; 0 otherwise. (WBES)

VARIABLES

❖ Firm-Level Variables:

Foreign Ownership (foreignown): A dummy variable equal to 1 if the firm is 100% foreign-owned.

Company Size: Categorical variables indicating firm size (e.g., small, medium, large).

Institution Age (institution_age_ln): The natural logarithm of the firm's age.

Access to Finance Obstacles (k30_major_obstacle): A dummy variable indicating whether the firm faces major obstacles in accessing finance (5 level of financial access).

Industry Dummies: Categorical variables for the firm's industry (manufacturing, services, core).

Website Presence (website): A dummy variable equal to 1 if the firm has a website.

VARIABLES

❖ National-Level Variables:

Average Price Gap: The difference between the national average gasoline price and a benchmark price. (interested variable)

$$\text{Average Price Gap} = \text{National Average Gasoline Price} - \text{Benchmark Price}$$

Average Price: The national average gasoline price. (interested variable)

Price Volatility: The fluctuation in gasoline prices.

GDP Per Capita: A measure of the economic status of the country.

Average Emission: A national-level indicator of environmental impact.

Squared_mean_price: squared value of the average price

I also explore **interaction terms** between the national-level variables (average_price_gap and price_volatility).

Distance from benchmark in the 1st half of 2015 (constant 2015 USD per litre)

Distance from benchmark in the 1st half of 2003 (constant 2015 USD per litre)

Gasoline consumption (thou. bbls per day)

- 100
- 1,000
- 10,000

MODEL

- A **Multilevel Logistic Regression Model** is employed to account for the hierarchical structure of the data, where firm-level observations are nested within countries and over time.
- The model includes **random effects for countries and fixed effects for years** to control for unobserved heterogeneity at the national level.
- I also explore **interaction terms** between the national-level variables (average_price_gap and price_volatility).

$$\text{logit}(P(Y_{ij} = 1)) = \beta_0 + \sum_{k=1}^K \beta_k X_{ijk} + u_j$$

Where:

- K is the number of individual-level predictors.
- X_{ijk} represents the k -th predictor for individual i in group j .
- β_k represents the coefficients for the individual-level predictors.
- $u_j \sim N(0, \sigma_u^2)$ is the random effect for group j , assumed to be normally distributed with mean 0 and variance σ_u^2 .

When including interaction terms between two variables, say X_1 (e.g., moderate obstacle) and X_2 (e.g., average price gap), the logistic regression formula can be expressed as:

$$\text{logit}(P(Y_{ij} = 1)) = \beta_0 + \beta_1 X_{1ij} + \beta_2 X_{2ij} + \beta_3 (X_{1ij} \times X_{2ij}) + u_j$$

Where:

- X_{1ij} and X_{2ij} represent the individual predictors for the interaction.
- β_3 is the coefficient for the interaction term ($X_{1ij} \times X_{2ij}$).
- u_j is the random effect for group j .





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ESTIMATION RESULTS

VARIABLES	Model 1	Model 2	Model 3	Model 4	Model 5
Average_price_gap	0.2594503	17.26483	17.01853	15.53625	15.39568
Average_price		-17.03195	-15.29216	-13.40654	-14.48365
Squared_mean_price			-0.6200524	-0.7423438	0.998128
Price_volatility				-1.462349	7.501717***
c.average_price_gap#c.price_volatility					-27.65894***
foreignown	0.1635668***	0.1634137***	0.1635759***	0.1635594***	0.1626452***
size_medium2099	0.0709169**	0.0708722**	0.0706445**	0.0705585**	0.0703233***
size_large100_and_over	0.0806708**	0.0807061**	0.0804936**	0.0803277**	0.0797765***
institution_age_ln	0.0595019**	0.0596638*	0.0597977*	0.0597391**	0.0598475***
k30_minor_obstacle	0.2640769***	0.2640456***	0.264068***	0.2640343***	0.2641014***
k30_moderate_obstacle	0.1315447***	0.1314399***	0.1315054***	0.131442***	0.1315908***
k30_major_obstacle	0.1427192***	0.1426652***	0.1424924***	0.1424522***	0.1431775***
k30_very_severe_obstacle	0.1955139***	0.1953379***	0.1951286***	0.1950763***	0.1954409***
a0_manufacturing	0.237253***	0.2374476***	0.2373026***	0.2373806***	0.2378081***
a0_services	0.1757199***	0.1759805***	0.176059***	0.1761628***	0.1760729***
website	0.4768597***	0.4766808***	0.4767712***	0.4767234***	0.4761997***
managerial_experience	0.0027101***	0.002691*	0.0026998*	0.0026853**	0.0025999***
r&d	1.501799***	1.501675***	1.501627***	1.501511***	1.500673***
formal_training	0.4253157***	0.4251626***	0.4252229***	0.4252541***	0.4252354***
gdp_percapita	0.000000148	-0.000000615	0.00000342	0.00000276	-0.00000979
average_emission	-0.0008786	-0.0009724	-0.0010655	-0.0011411	-0.0012418
RANDOM					
Intercept	0.5617	0.5866477	0.5542963	0.5345724	0.3387384
No groups	47	47	47	47	47
N	31,797	31,797	31,797	31,797	31,797
LR test	926.1	922.72	910.78	923.37	667.02
Year	Yes	Yes	Yes	Yes	Yes

FINDINGS AND CONCLUSIONS:



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- **High Taxation (Positive average_price_gap):** When fuel prices are heavily taxed, it sends a signal to reduce consumption or shift towards alternatives. However, in a highly volatile market (with unpredictable and fluctuating prices), this policy signal is less effective. The uncertainty brought by price volatility could make consumers and firms hesitant to change their behavior (e.g., investing in energy-efficient technologies) despite high taxes.

- **Example:** In a volatile market, even if fuel price is heavily taxed, businesses might delay investments in alternative energy or efficiency upgrades because they are unsure how long the high prices will persist.

- **Subsidies (Negative average_price_gap):** When oil prices are subsidized, it usually encourages consumption. However, if the market experiences high volatility, subsidies might not be as effective in boosting consumption or economic activity. Firms and consumers may be more cautious, fearing that volatile prices could negate the benefits of current subsidies.

- **Example:** A government may subsidize fuel prices to support the economy, but if price volatility is high, consumers may still be reluctant to spend more, and firms might hesitate to make long-term investments based on these lower prices.

Policy Implications:

- **High Volatility Reduces Policy Effectiveness:** The negative interaction indicates that price volatility undermines the intended effects of both taxes and subsidies on oil. In stable markets, taxation and subsidies have clearer effects on consumption and investment behaviors. In contrast, high volatility makes firms and consumers more risk-averse, reducing their responsiveness to these policy measures.

- **Stabilizing Markets:** For policies such as taxation or subsidies to be more effective, governments and regulators might need to focus on reducing market volatility. A stable market environment allows economic agents to better assess and react to policy signals.



THANK YOU FOR YOUR ATTENTION

ANY QUESTIONS?



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